👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 7 Using the Repair Manual (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Week 5: Manual Reading Handout (Single Cylinder L-Head repair manual), iCADEMY Bronze 1 Warranty Policy, Bronze 2 Product Liability, Bronze 3 Parts Look-Up, Tech bits Digital workshop manuals”)	**Level 2 piece**
Manual Reading Handout (B&S L-Head manual)	Notes + Manual Look-Up lab packet; use a 1-page 'manual map' and staff-guided look-ups
iCADEMY Bronze 1 Warranty, Bronze 2 Product Liability	Slides 'Warranty vs Product Liability' (our own content)
iCADEMY Bronze 3 Parts Look-Up, Digital workshop manuals	Slides 'Parts Look-Up' + lab packet
Class slides	Deck: 15 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 10 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no “all of the above”
- Staff mark the manual's sections with sticky tabs before class
- L1/L2: staff find the page; student finds the number on that page
- Never copy manual pages onto the site (copyrighted): use the class copies

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part

4. What to grade

- Manual Reading Handout: L1 answers 3 look-ups with staff; L2 answers 5-6; L3 all with the manual map
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Repair manual**: The Briggs & Stratton manual is copyrighted and stays in the classroom; our materials teach how to use it without copying it.
- **iCADEMY Bronze 1-3**: STIHL iCADEMY is login-only; we wrote our own warranty/liability/parts content.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Repair Manual Look-Up (Level 2)

Small Engines Overhaul · Unit 7 · goes with the class “Manual Reading Handout” (handout points). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Manual Reading Handout”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (handout points)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: Spec values depend on the class manual and engine models; our key shows typical values only. The manual itself stays in the classroom (copyrighted).

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

📘 Repair Manual Notes

Small Engines Overhaul · Unit 7 Using the Repair Manual · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Finding your way

1. Match the manual part to what it does:

Table of contents	specs by model
Index	all parts with numbers
Chart	sections in order
Exploded diagram	topics A to Z

B. Specs and parts

2. Find your _____ number before using a chart.
3. Order parts using the _____ number.
4. Do not _____ in the shop manuals.

Word bank: **model** **part** **write**

C. Warranty and liability

5. A _____ is the maker's promise to fix defects.
6. Using wrong parts can _____ a warranty.
7. Shops follow the manual to avoid product _____.

Word bank: **liability** **void** **warranty**

Level 2 version of the gen ed “Manual Reading Handout.”

📖 Using the Repair Manual: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Model number

The number on the engine (often stamped on the shroud or block) that tells which manual pages and parts fit it.



Repair manual

The maker's book with steps, specs and charts for repairing an engine.



Index

The alphabetical list at the back or front that tells which section and page a topic is on.



Section

A chapter of the manual about one system, like ignition or carburetion.



Chart

A table in the manual that gives specs by model number.



Part number

The number used to look up and order the exact replacement part.



Warranty

The manufacturer's promise to repair or replace a product for a set time if it fails from a defect.



Product liability

The legal responsibility of makers and repair shops if a product causes harm; why shops follow the manual.

📘 Repair Manual Look-Up

Small Engines Overhaul · Unit 7 · Level 2



Name: _____

Modified from the class “Manual Reading Handout”.

🧺 Supplies

- ☐ 📘 **Class repair manual** classroom copy, no writing
- ☐ 🗺️ **Manual map card** staff-made sticky tabs
- ☐ 💻 **Computer** parts look-up, with adult
- ☐ 🏷️ **Engine with model tag**
- ☐ 📋 **Look-up sheet**

📘 Steps: check each box when you finish

1. 🏷️ Record the model, type and code numbers. ☐
2. 🔤 Use the index to find 'spark plug gap'; record section and page. ☐
3. 📊 Read the chart row for your model; record the spec and units. ☐
4. 🔤 Repeat for 'armature air gap' and 'valve clearance'. ☐
5. 🖼️ Find the air filter in an exploded diagram; record its reference number. ☐
6. 💻 With an adult, look up the part number online. ☐
7. 📘 Return the manual; no marks in it. ☐

🚧 **Under construction:** Spec values depend on the class manual and engine models; our key shows typical values only. The manual itself stays in the classroom (copyrighted). *Check with the class teacher.*

📘 Repair Manual Look-Up · Prep Sheet

Small Engines Overhaul · Unit 7 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🔧 **Mechanic**	Does the main engine job with the tools	
🧰 **Tool helper**	Gets tools from the board and returns them to staff	
📋 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Wipes the bench, sweeps, checks tools are back	

🧰 Tools and supplies: check when you have it

- ☐ 📘 **Class repair manual** classroom copy, no writing
- ☐ 🏷️ **Engine with model tag**
- ☐ 🗺️ **Manual map card** staff-made sticky tabs
- ☐ 📋 **Look-up sheet**
- ☐ 💻 **Computer** parts look-up, with adult

⚠️ Safety and manners focus

🥽 **Safety glasses** on the whole time.

👖 **Coveralls** and closed-toe shoes; tie back long hair.

🔧 Use tools **with adult supervision**. Right tool for the job.

📘 Manuals are shared: **no writing**, clean hands.

🧼 **Clean up** and wash hands when you finish.

✅ Before you start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

📘 Repair Manual Look-Up · Reflection

Small Engines Overhaul · Unit 7 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🥽 Wore safety glasses and used tools safely

☐ 🔧 Finished the Repair Manual Look-Up steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about engines?

1. _____
2. _____

Ideas: **The engine needs…** **I saw…** **Next time…**

✅ Using the Repair Manual Quiz

Small Engines Overhaul · Unit 7 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🏷️ The first thing to find is the…

A. Price | B. Model number | C. Color

2. 🔤 The index lists topics…

A. By color | B. By price | C. Alphabetically with pages

3. 📊 Spec charts are read by…

A. Model number row | B. Page color | C. Guessing

4. 📑 A section of the manual covers…

A. The whole world | B. Only prices | C. One system, like ignition

5. 🖼️ An exploded diagram shows…

A. Parts spread out with numbers | B. A broken engine | C. Only the box

6. 🔩 Parts are ordered by…

A. Color | B. Part number | C. Size guess

7. 🛡️ A warranty…

A. Covers everything forever | B. Covers defects for a set time | C. Is a tool

8. ❌ A warranty can be voided by…

A. Misuse or wrong parts | B. Reading the manual | C. Using fresh gas

9. ⚖️ Product liability is about…

A. Engine speed | B. Fuel mix | C. Responsibility if a product causes harm

10. 🚫 In shop manuals you…

A. Never write | B. Highlight | C. Tear pages

📘 Repair Manual Notes · ANSWER KEY

Small Engines Overhaul · Unit 7 Using the Repair Manual · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Finding your way

1. Match the manual part to what it does:

(Key: Table of contents → sections in order · Index → topics A to Z · Chart → specs by model · Exploded diagram → all parts with numbers)

Table of contents	specs by model
Index	all parts with numbers
Chart	sections in order
Exploded diagram	topics A to Z

B. Specs and parts

2. Find your **model** number before using a chart.
3. Order parts using the **part** number.
4. Do not **write** in the shop manuals.

Word bank: **model** · **part** · **write**

C. Warranty and liability

5. A **warranty** is the maker's promise to fix defects.
6. Using wrong parts can **void** a warranty.
7. Shops follow the manual to avoid product **liability**.

Word bank: **liability** · **void** · **warranty**

Level 2 version of the gen ed "Manual Reading Handout."
🔑 Key written by our team from the class teacher's materials, please check.

✅ Using the Repair Manual Quiz · ANSWER KEY

Small Engines Overhaul · Unit 7 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🏷️ The first thing to find is the… Manual Handout

A. Price | **B. Model number** | C. Color

2. 🔤 The index lists topics… Manual Handout

A. By color | B. By price | **C. Alphabetically with pages**

3. 📊 Spec charts are read by… Manual Handout

A. Model number row | B. Page color | C. Guessing

4. 📑 A section of the manual covers… Manual Handout

A. The whole world | B. Only prices | **C. One system, like ignition**

5. 🖼️ An exploded diagram shows… Parts Look-Up

A. Parts spread out with numbers | B. A broken engine | C. Only the box

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A. Color | **B. Part number** | C. Size guess

7. 🛡️ A warranty… Warranty

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8. ❌ A warranty can be voided by… Warranty

A. Misuse or wrong parts | B. Reading the manual | C. Using fresh gas

9. ⚖️ Product liability is about… Liability

A. Engine speed | B. Fuel mix | **C. Responsibility if a product causes harm**

10. 🚫 In shop manuals you… Manual Handout

A. Never write | B. Highlight | C. Tear pages

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “Manual Reading Handout”. Score: __ / 10. 🔑 Key written by our team from Mr. Davis's materials, please check.